Using deep learning to generate synthetic tax records

John Ricco

The Budget Lab at Yale

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What do we want?

Augmented tax unit microdata
for every year, including future years

What do we have?

- **One year of tax data, filers only**
  - About ten years out of date
  - Some values are blurred and some records are distorted
  - No non-filers, and none of the other variables a tax calculation needs: consumption, wealth, unrealized gains, estates
- **Surveys** that hold some of this information, but noisily and not at administrative quality
- **Aggregate targets**
  - Past years: IRS totals by AGI group, filing status and age range
  - Future years: CBO-style projections of 1040 items, or of total wages, business profits and similar totals

The task

Turn these scattered sources into

synthetic, realistic, augmented tax unit microdata

for every year

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m

1. **Start with the IRS PUF**, whatever year is available.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Billionaires	Married	1	$1.2m	$14m	$310m
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m
Non-filers	Single	2,310	$0	$0	$0
	Single	1,870	$6.5k	$0	$0

2. **Add records**: non-filers, and the very rich the PUF leaves out.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	812	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,404	$38k	$0	$0	$9k	$34k	$6.1k
	Married	655	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,120	$51k	$12	$0	$31k	$47k	$0
	Single	3	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,310	$0	$0	$0	$2k	$19k	$0
	Single	1,870	$6.5k	$0	$0	$1k	$15k	$1.2k

3. Add columns: estimate Y given X in a survey, where X is also on the PUF; draw Y for each record.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	860 ↑	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,290 ↓	$38k	$0	$0	$9k	$34k	$6.1k
	Married	702 ↑	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,180 ↑	$51k	$12	$0	$31k	$47k	$0
	Single	2 ↓	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,450 ↑	$0	$0	$0	$2k	$19k	$0
	Single	1,760 ↓	$6.5k	$0	$0	$1k	$15k	$1.2k

4. **Reweight to the target year**: grow weights for population and dollars for prices, then move each weight up or down until the year's totals are hit.

Made-up records.

What were the problems with this approach?

- **It did not work well for past years.** The affordability project made this clear.
- **A separate model for each variable**, each estimating its conditional relationships in its own way.
- **Reweighting worked for only a few past years.** Tax law changed what the IRS targets measure, mainly TCJA cutting back itemized deductions.
- **Stretched further every year.** Plausible records for future years, but worse with each year away from 2015.

The ideal approach

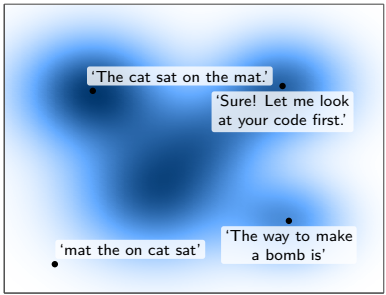
1. Take in survey data covering the joint relationships among every variable we might want to model.
2. Learn all of those relationships in **One Huge Joint Probability Distribution**.
3. For any target year, supply the targets that describe that year, and pull out of it the records that match them.

The result: a realistic synthetic microdata set for that year.

From a conversation with John and Harris.

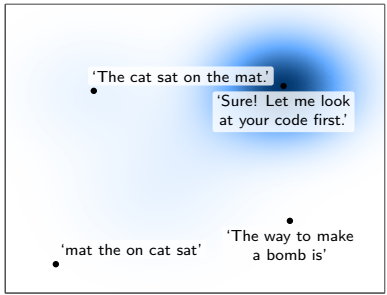
Language models: pretraining, then post-training

Pretraining



Read the whole internet and learn which sentences are likely: a prior over language

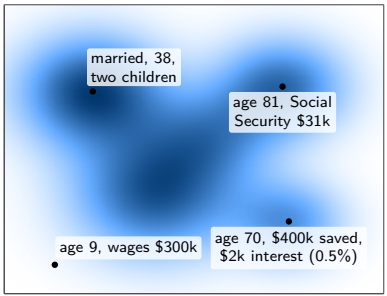
Post-training



Squeeze the mass toward a helpful, harmless assistant, and away from nonsense and harm

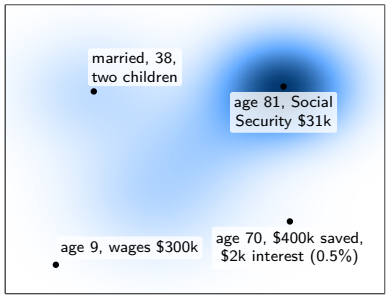
Our model: pretraining, then post-training

Pretraining



Learn which records are likely, from every source: a prior over records

Post-training for 2040



Squeeze the mass toward records that hit 2040's totals, each still coherent

Post-training finds the distribution closest to the prior that hits the year's totals.

Writing one record, one item at a time

The record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile

We set four things before any draw:

- the source the record is written as (the PUF),
- its year (2015),
- that it is a filer,
- its AGI group, one of 12.

Then we ask the model for the next item.

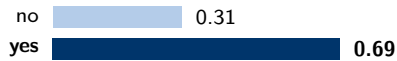
Made-up record and probabilities.

Writing one record, one item at a time

The record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes

Next: married?



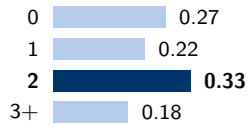
Made-up record and probabilities.

Writing one record, one item at a time

The record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2

Next: dependents?



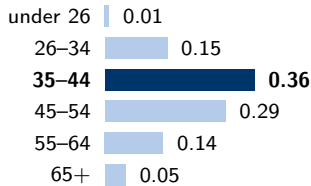
Made-up record and probabilities.

Writing one record, one item at a time

The record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44

Next: age range?



Made-up record and probabilities.

Writing one record, one item at a time

The record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44
filing status	joint
credit children	2 and 2

Then filing status and the number of child credit and EITC children, drawn the same way.

Some draws follow rules: a married unit is always a joint return, as the PUF codes it.

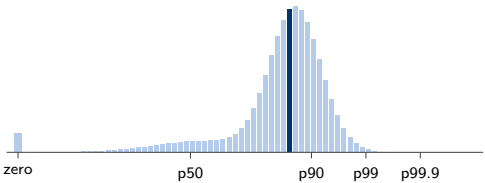
Made-up record and probabilities.

Writing one record, one item at a time

The record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44
filing status	joint
credit children	2 and 2
wages	82nd–84th pct. band

Next: wages?



One bar for zero and 73 for positive wages, cut at percentiles of wage earners. Drawn (dark): the band from the 82nd to the 84th percentile.

Made-up record and probabilities.

Writing one record, one item at a time

The record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44
filing status	joint
credit children	2 and 2
wages	82nd–84th pct. band
interest	zero
dividends	40th–42nd pct. band
Schedule C	zero
···	about 90 tax items

Made-up record and probabilities.

Every other tax item the same way, in a random order: each draw given everything drawn so far.

Totals are never drawn. Capital gains, business income, ordinary dividends and the adjustments are added up from their drawn parts.

Next: the items the PUF does not have.

The rest of the record: what each survey would have recorded

Then the model is asked about the same unit as each survey would see it: the source label is switched, and the items that survey lacks are hidden from the model.

Asked as	Draws	Sees, besides this phase's draws so far
CPS 2019	exact ages of filer and spouse	the items the CPS shares with the PUF
SCF, its wave	20 balance-sheet items, unrealized gains	income items the SCF shares, exact ages
CEX 2019	8 consumption categories	married, dependents, ages, wages, Social Security, self-employment income
CPS 2019	sex, dependents' ages, spouse's share of wages, weeks and hours, childcare	the items the CPS shares with the PUF, exact ages
SIPP 2019	each adult's tips, as a share of wages	married, dependents, ages, wages, spouse's share

- Each record's SCF wave is drawn at random: 2019 (55%), 2022 (33%), 2016 (12%).
- Rules: mortgage interest on the return means a home and a mortgage; a dependent filer gets no survey items.

What one step returns, and how it becomes dollars

A probability for every possible value

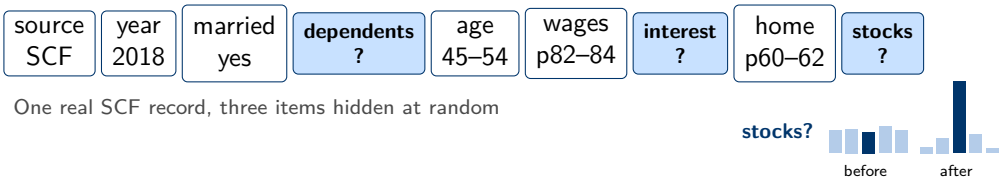
- A category (married, dependents 0–3, age range): one per value.
- An amount: one for each of 84 bands.
  - exactly zero
  - 10 bands of losses
  - 73 bands of positive amounts, by percentile among holders: 2 points wide up to the 90th, then 1, 0.1 and 0.01 at the very top
- No functional form: a spike at zero or a long top tail costs nothing extra.

Example dollar amounts are made up.

Bands become dollars at the end

- Each band becomes one real value from that band, picked at random: from 2015 PUF returns for tax items, the record's SCF wave for wealth, the 2019 CEX for spending.
- Then grown to 2019 item by item (tax items: SOI's growth in dollars per return with the item).
- Example: wages band 82nd–84th percentile; a real 2015 wage in it, \$96,400; grown to 2019, \$108,000.

Where the probabilities come from: training



1. On each real record, from all five sources, hide anywhere from one to all of its items. Source and year stay shown.
2. The model returns probabilities for each hidden item.
3. Adjust its 1.4 million parameters to raise the probability on the true values (maximum likelihood, by survey weight; top records count extra). 80 passes over all records.

Any set can be hidden, so one model answers "this item, given any others": what each step of writing a record asks.

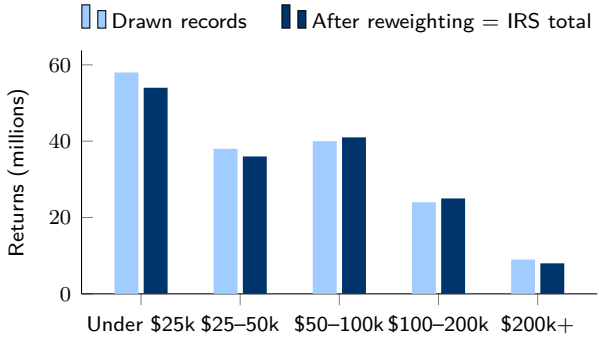
Made-up record and probabilities. The dark bar is the record's true band.

Post-training: reweight to hit the totals

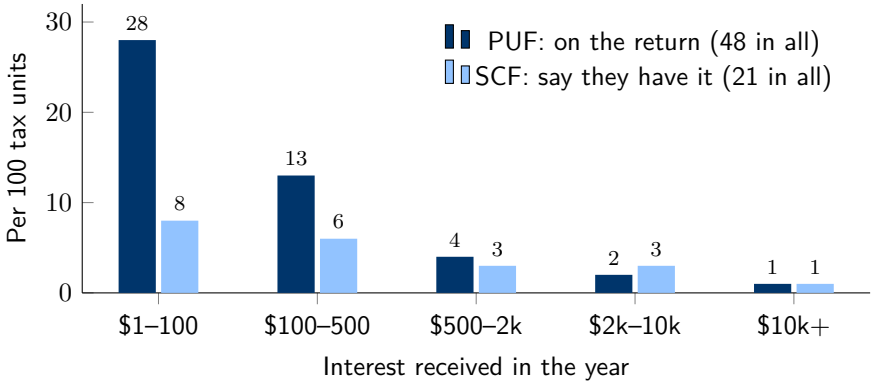
1. Draw 400,000 filers, a set number per AGI group, and non-filers as the CPS 2019 would see them.
2. Start each weight at its AGI group's 2015 returns over the records drawn in it.
3. Move the weights as little as possible (as in entropy balancing) until each of 2019's 753 targets is within one standard error (SCF targets: half).

The records do not change, only their weights.

Made-up numbers in the chart.



Challenge: surveys and returns measure differently



The gap is all in small amounts: they go unreported in the survey. Above $2,000 the two sources agree.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted.

The fix: two slots and a fill-in

SCF household	interest, as said	interest, as on a return
household 1	$250	$250
household 2	$0	?
household 3	$0	?
household 4	$1,800	$1,800

The answer given is kept. A second slot holds what the return would show, the only slot the PUF has.

Per 100 couples	PUF	SCF	Fill in
$1–100	28	8	20
$100–500	13	6	7
$500–2k	4	3	2
$2k and over	3	3	0
Any	48	21	28

The PUF sets how many are filled in, and at what sizes. The rest of the SCF record sets which households.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted. Rows may not add because of rounding.

Other challenges

- **Building tax units from surveys.** Surveys count households. Each one is split into tax units by the same rules for every survey, and then we decide who files.
- **The return itself changes.** TCJA cut back itemizing; new items, like the tips and overtime deductions, are on no past return.
- **A thin top.** Few very rich records in any source.
- **Future years.** No microdata, only a handful of projected totals.
- **Checking the result.** No source has every variable, so some relationships have nothing to check against.

Where we are

- A 2019 file: 400,000 filer records, plus non-filers
- Five sources: the 2015 PUF, the SCF, the CEX, the CPS and the SIPP, every survey split into tax units by the same rules
- Every PUF item, plus a balance sheet, consumption, tips and auto loan interest
- About 750 targets, all met within their standard errors: IRS tables by AGI, filing status and age, the top 1%, 0.1% and 0.01%, W-2 tips, SCF wealth

What is left

- Rules applied after drawing: credit eligibility, death probabilities, unrealized gains, the basis of capital gains
- The file Tax-Simulator reads, and a first run of Tax-Simulator on it
- A tax calculator inside post-training, for targets on computed tax
- Other years: 2015 to 2023, then the projection years
- The very top: the PUF's aggregate records and the Forbes 400

Questions